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# SEEKING SCIENCE

## MONTHLY NEWSLETTER

STEM Action Teen Institution



**An Examination of  
Magnetoreception in  
the Animal Kingdom**

Kenny Wu

**Physic of Formula  
One**

Richard Wang

**The Exceptional  
Resilience of the  
Phylum Tardigrada**

Owen Chen

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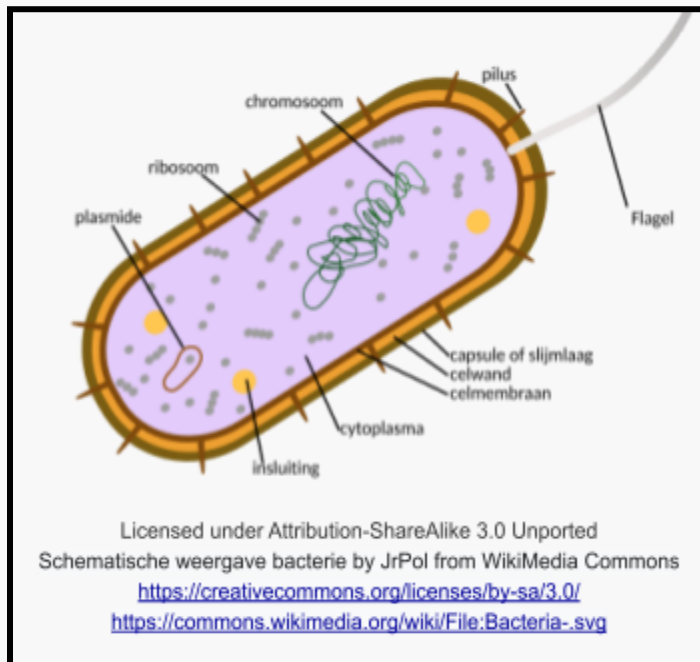
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# Plastic-Eating Bacteria

By Emily Ma

Plastic-eating bacteria are microorganisms capable of breaking down and consuming plastic. These bacteria can play a significant role in reducing plastic pollution and environmental contamination. Plastic-eating bacteria, such as *Ideonella sakaiensis*, *Comamonas testosteroni*, and *Pseudomonas aeruginosa*, have shown the ability to degrade various types of plastic, including PET, polyethylene, polypropylene, and



polystyrene. These bacteria produce enzymes that break down plastic into smaller molecules, which they then consume as a source of carbon and energy by targeting the chemical bonds between the plastic's monomers. Their potential applications include bioremediation, where they can help clean up plastic pollution in landfills, oceans, and other environments, as well as plastic recycling by converting plastic waste into valuable materials

like monomers or biofuels. Additionally, they could contribute to sustainable manufacturing by enabling the development of biodegradable plastics. However, challenges remain, such as the slow degradation rate, limited effectiveness against certain plastics, and environmental factors like temperature and pH that can influence their activity. Despite these limitations, plastic-eating bacteria offer promise as a potential solution to plastic pollution, and ongoing research aims to improve their efficiency and

expand their capabilities, contributing to environmental sustainability by reducing plastic waste and its harmful impact on the planet.

Genetically modified plastic-eating bacteria offer a promising solution to reduce plastic pollution, but their introduction into the environment raises concerns about unintended ecological consequences. Factors such as temperature, pH, and interactions with other microorganisms can significantly influence their efficiency. Ongoing research focuses on enhancing the plastic-degrading capabilities of these bacteria through genetic engineering, optimizing their enzymes to function more effectively across a wider range of plastics. Scientists are also exploring ways to incorporate plastic-eating bacteria into waste management systems by integrating them with mechanical and chemical recycling methods, creating a more comprehensive approach to addressing plastic waste. While these bacteria show immense potential in bioremediation, recycling, composting, and bioplastic production, challenges such as slow degradation rates, limited plastic specificity, and possible ecological impacts must be addressed to improve their effectiveness.

With continued advancements in genetic engineering and biotechnology, plastic-eating bacteria could become a powerful tool in combating plastic pollution and promoting environmental sustainability. Overcoming these challenges will help unlock their full potential, paving the way for a cleaner and more sustainable future.

# Biochemistry

By Audrey Wu

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Biochemistry is the study of chemicals and reactions inside living organisms. It combines biology and chemistry to understand how molecules like proteins, nucleic acids, lipids, and carbohydrates help cells function and stay organized. Biochemists explore processes such as metabolism, enzyme function, signal transmission, and gene expression. This knowledge is crucial for medicine, agriculture, and environmental science, aiding in disease treatment and food production.

A key focus of biochemistry is on macromolecules—large molecules essential to cell structure and function. Proteins work as enzymes, hormones, and structural components, performing most of the cell's activities. Nucleic acids like DNA and RNA store and transfer genetic information, ensuring traits are passed down. Lipids construct cell membranes and act as energy storage, while carbohydrates provide quick energy for cellular tasks. Understanding these interactions helps biochemists reveal how cells are organized and function.

Biochemistry has made significant advances in medicine. By examining biochemical processes related to diseases like cancer, diabetes, and Alzheimer's, scientists identify where drugs can be effective and develop treatments. Enzyme inhibitors have been crucial in treating infections and chronic conditions. Progress in genetic engineering has led to genetically modified organisms (GMOs), gene therapies, and personalized medicine, offering solutions to some of the world's toughest health issues.

In the future, biochemistry will have even more breakthroughs. Technology like high-speed DNA sequencing and modeling allows detailed study of biological processes. The field is expanding into synthetic biology and biotechnology, providing new solutions for renewable energy, environmental sustainability, and food supply. Continuous research and innovation in biochemistry will keep it central to modern science, influencing future healthcare and even agriculture.

# Physics in Running: How Different Shoes Impact Speed

By Brandon Wang

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The type of running shoe can impact a runner's speed because of the weight, cushioning, grip, and biomechanics. Running shoe technology has transformed competitive running throughout the years, with top marathoners and sprinters counting on specialized shoes to maximize efficiency.

Energy return is one of the key elements in a fast-running shoe. New "super shoes" feature carbon fiber plates and very responsive foams (like Nike's ZoomX or Adidas's Lightstrike Pro) that store energy when the foot hits the ground and return it at toe-off. This aids in propulsion, decreases fatigue, and improves running economy. Research has demonstrated that these shoes can make running 4% or more efficient, resulting in quicker race times. On the other hand, minimalist shoes, with little or no cushioning, promote a more natural running stride but cause greater stress on foot muscles and are, therefore, less ideal for maximum speed over extended distances. Another important consideration is shoe weight. Lighter shoes decrease the energy it takes to lift the foot with every stride, enabling runners to run more quickly. Racing flats and track spikes are designed with this in mind, weighing in at as little as 100-150 grams per shoe. For sprinters, track spikes are an even bigger player in speed optimization. Their rigid plates and strategically placed spikes increase traction and force application, allowing forceful push-offs and greater acceleration. For runners covering long distances, cushioning and stability can affect efficiency. Stability shoes can alter the stride and eliminate wasted energy for overpronators—runners who have excessively rolling inward feet. Neutral runners can go

for lightweight, flexible shoes to provide a freer gait cycle, optimizing their speed without additional limitation.

Finally, the most important factor in off-road racing is traction. Trail running shoes are more aggressively treated to have better grip on uneven terrain and to avoid slipping that would decelerate a runner. Road racing shoes favor the smooth outsole and springy midsole for optimum performance on the road. In general, the best speed shoe is based on race distance, running surface, and personal biomechanics. With carbon-plated marathon shoes, minimalist track spikes, and specialty trail runners, shoe technology continues to advance, stretching the limits of human performance.

# Physics of Formula One

Richard Wang

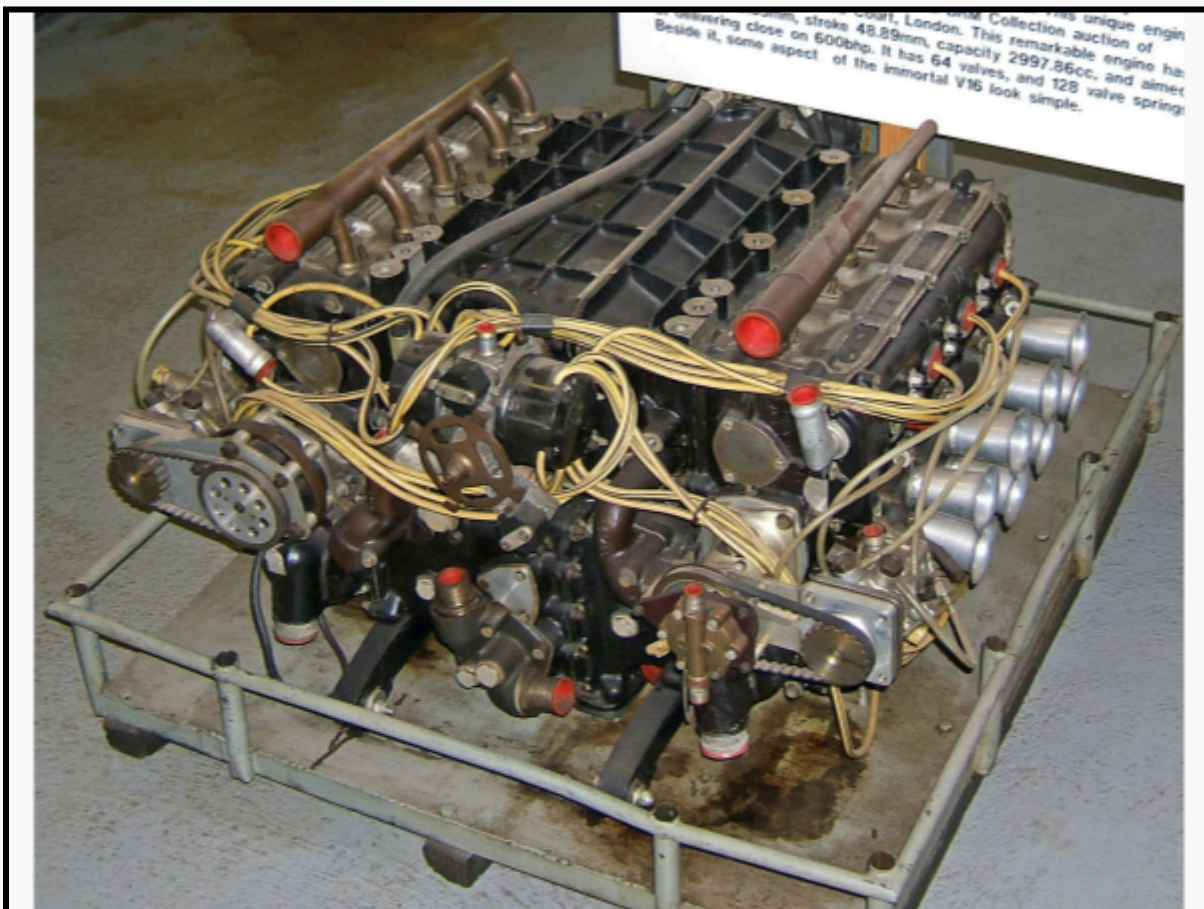
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Every year, there is a new Formula One season where racing teams design and create the best cars for ultimately winning the world championship. Formula 1, also known as F1, is the highest class of international racing, and it is quite fascinating how teams push the limits of physics and engineering in designing cars that are not only fast but also agile. Cars can achieve speeds of up to 220 miles per hour and are capable of turning corners at tremendous speeds. These are all made possible by important factors such as aerodynamics, engine power, and engineering techniques.

Something almost as important as the engine behind an F1 car is the aerodynamics. Aerodynamics aim to reduce drag, create downforce, and minimize lift. As cars move at high speeds, air resistance is opposing their motion, so the car's body is optimized to counteract this force; this includes things like smoothing out the body of the car, reducing sharp edges, and shifting the area of the car away from the front. In addition to drag, F1 cars face an upward force called lift. F1 cars have wings in the front and back that create downforce, which can counteract that lift. This downforce pushes the car into the ground, which improves its grip and stability. This is very important for turning corners, as this generated downforce allows F1 cars to turn corners at higher speeds without slipping. Aerodynamics plays a great role in allowing cars to gain faster lap times.

The engine is where the power of a Formula car is generated. The formula in F1 means that there are set rules about the car, which includes the engine. The engine of an F1 car contains a turbocharger. The turbochargers work by forcing more air into the engine, resulting in increased power output. Gases spin a turbine, which powers a compressor that pushes air in. F1 engines have a limit of 15,000 rotations per minute and produce a power output of nearly 1000 horsepower. There are also electrical

components to the engine. Hybrid engines have electric motors powered by batteries. These electric motors add power to the crankshaft. This electrical component is important because it allows the car to recover energy when braking. This technology, called the kinetic energy recovery system, enables the car to use that recovered energy to boost engine performance.



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# The Intriguing Journey of the Collatz Conjecture

By Aidan Hong

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The Collatz Conjecture stands as a prominent unsolved problem within the field of mathematics, specifically in the domain of number theory. Proposed by Lothar Collatz in the early twentieth century, this conjecture concerns a seemingly elementary iterative process defined for all positive integer values.

The process is governed by a conditional rule set. For any given positive integer, if the integer is even, the subsequent term in the sequence is obtained by dividing the integer by two. Conversely, if the integer is odd, the subsequent term is derived by multiplying the integer by three and then adding one to the product. The Collatz Conjecture posits that the repeated application of these rules, commencing from any positive integer, will inevitably lead to the value of one.

Consider the integer eleven as an illustrative example. As eleven is an odd number, the next term in its Collatz sequence is determined by the operation three multiplied by eleven, plus one, resulting in thirty-four. The integer thirty-four is even, and thus the subsequent term is thirty-four divided by two, yielding seventeen. Continuing this iterative process, seventeen (odd) leads to three multiplied by seventeen, plus one, equaling fifty-two. The sequence progresses: fifty-two (even) becomes twenty-six, which in turn becomes thirteen (even). Thirteen (odd) yields three multiplied by thirteen, plus one, or forty. The sequence continues through twenty, ten, five, sixteen, eight, four, two, and finally reaches one. This empirical observation, replicated across a vast number of initial integers, lends credence to the conjecture.

Despite the compelling empirical evidence gathered through extensive computational verification, a rigorous mathematical proof of the Collatz Conjecture remains outstanding. Modern computing power has enabled the testing of the conjecture for extraordinarily large initial values, with no counterexamples having been discovered to date. However, in the axiomatic framework of mathematics, empirical validation, regardless of its scale, does not constitute a formal proof applicable to all positive integers. The possibility remains that an as-yet untested integer exists whose Collatz sequence does not ultimately reach one, either by entering a cycle distinct from the one containing one, or by diverging to infinity.

The inherent difficulty in establishing the truth of the Collatz Conjecture underscores the potential for intricate behavior to emerge from simple deterministic rules. It serves as a compelling example of a problem in number theory that is easily stated and understood, yet profoundly challenging to resolve. The conjecture continues to motivate mathematical research, with investigators employing a variety of analytical techniques in pursuit of either a proof of its universality or the identification of a counterexample.

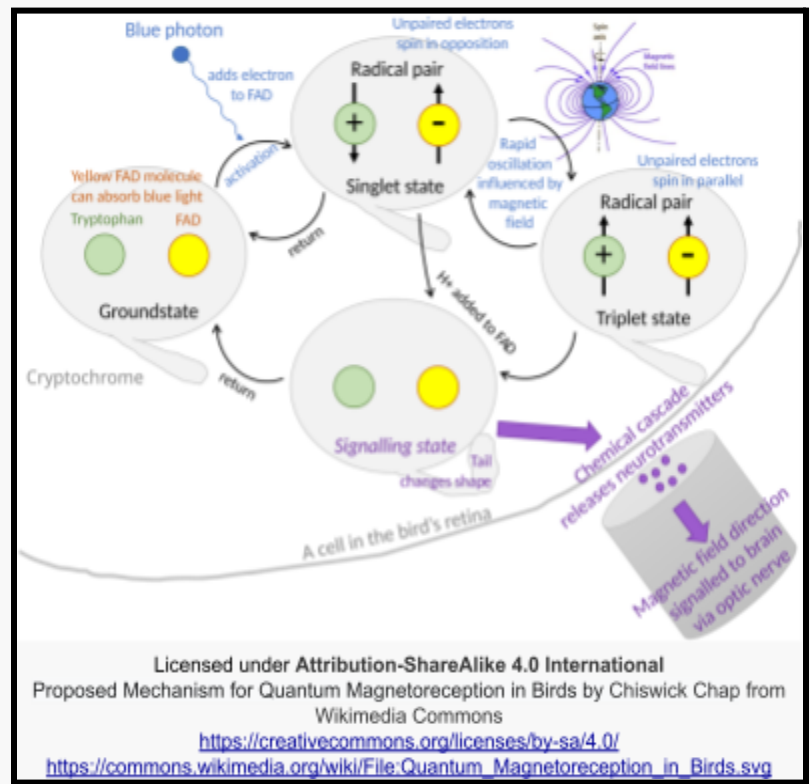
In summary, the Collatz Conjecture represents a significant open question in mathematics. Its assertion that all positive integers, when subjected to a specific iterative process, will eventually reach one remains unproven. The conjecture's simplicity contrasts sharply with the complexity of the attempts to resolve it, highlighting the enduring mysteries that reside within the seemingly elementary structure of the integers.

# An Examination of Magnetoreception in the Animal Kingdom

By Kenny Wu

Magnetoreception, the capacity of certain organisms to perceive the Earth's magnetic field, represents a fascinating area of sensory biology. This innate ability, distinct from the more commonly understood senses, allows a diverse array of animal species to orient themselves within their environment and navigate across significant distances. While the precise mechanisms underlying magnetoreception are still under investigation, the evidence for its existence and its ecological significance continues to expand.

The phenomenon of magnetoreception has been documented across a wide phylogenetic spectrum. Avian species, particularly migratory birds, have long been recognized for their remarkable navigational feats. Research suggests that these animals utilize the Earth's magnetic field as one of several cues to maintain their migratory routes, especially over unfamiliar terrain. Similarly, marine animals, such as



sea turtles and salmon, exhibit an impressive ability to return to specific natal sites after traversing vast oceanic distances. Magnetoreception is hypothesized to play a crucial role in this homing behavior, providing a directional sense within the expansive marine environment.

Beyond these well-known examples, evidence for magnetoreception has emerged in various other taxa. Insects, including monarch butterflies undertaking transcontinental migrations, and certain species of bees, appear to employ magnetic field detection for orientation purposes. Furthermore, investigations into mammalian behavior have yielded intriguing results, with some studies suggesting a magnetic sensitivity in bats and potentially even in domestic animals. The widespread occurrence of this sensory modality underscores its potential evolutionary advantages for spatial awareness and long-distance travel.

The physiological basis of magnetoreception is a subject of ongoing scientific inquiry. Two primary hypotheses have garnered significant attention. The first involves the presence of magnetosensitive cells containing biogenic magnetite, a naturally occurring magnetic mineral. These cells, often located in structures such as the beak of birds or the olfactory epithelium of other animals, are theorized to respond to the Earth's magnetic field, transmitting neural signals that provide directional information. The second prominent hypothesis centers on light-dependent magnetoreception, potentially mediated by specialized photoreceptor proteins within the eye. These proteins are proposed to be sensitive to the influence of magnetic fields on photochemical reactions, allowing animals to effectively "see" the magnetic field as a visual cue, particularly under specific light conditions. It is plausible that different animal groups, or even different contexts within the same species, may utilize one or both of these mechanisms.

The ecological implications of magnetoreception are considerable. For migratory species, the ability to accurately perceive and respond to the Earth's magnetic field is likely crucial for successful navigation, enabling them to reach breeding and foraging

grounds across continents and oceans. This innate compass sense can be particularly important in the absence of other reliable cues, such as celestial landmarks or familiar terrain. Moreover, for animals with more localized ranges, magnetoreception may contribute to spatial orientation within their immediate environment, aiding in tasks such as foraging and homing.

In conclusion, magnetoreception represents a sophisticated sensory adaptation that allows a diverse array of animals to interact with the Earth's magnetic field. While the precise sensory mechanisms continue to be elucidated, the evidence strongly suggests that this ability plays a vital role in the orientation and navigation of numerous species across terrestrial and aquatic environments. Further research into the neurobiological and behavioral aspects of magnetoreception promises to yield deeper insights into the sensory world of animals and the remarkable adaptations that facilitate their survival and movement within the global ecosystem.



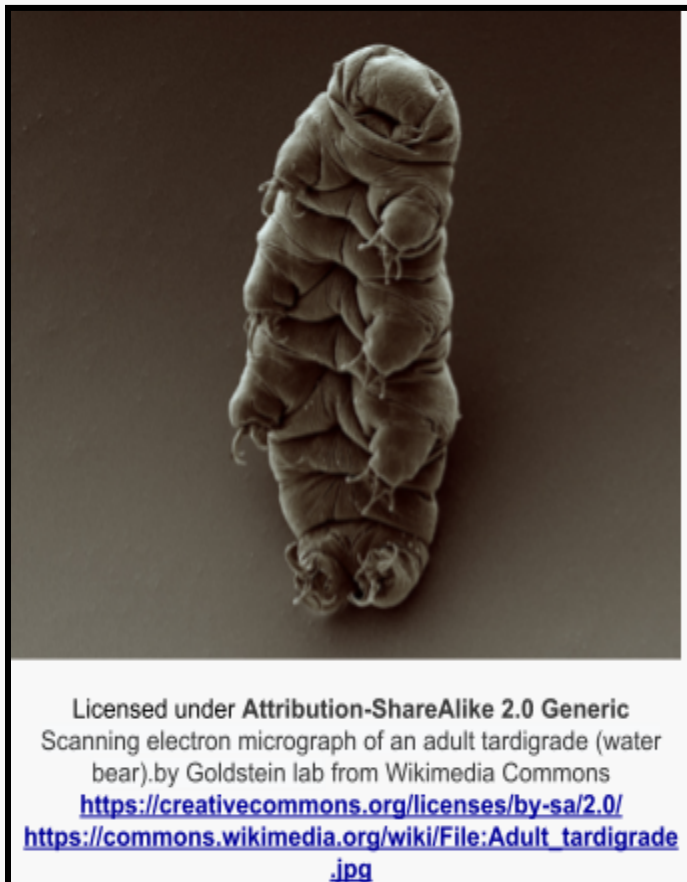
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# The Exceptional Resilience of the Phylum Tardigrada

By Owen Chen

The phylum Tardigrada encompasses a group of microscopic invertebrates renowned for their extraordinary capacity to survive a wide array of extreme environmental conditions. Commonly referred to as tardigrades, water bears, or moss piglets, these diminutive metazoans exhibit a global distribution, inhabiting diverse ecological niches ranging from high-altitude terrestrial environments to the deep sea. Their remarkable tolerance to stressors such as extreme temperatures, pressures, radiation, dehydration, and chemical exposure has positioned tardigrades as a subject of significant interest in biological research.



Tardigrades are typically less than one millimeter in length and possess a segmented body plan characterized by four pairs of lobopodia legs terminating in claws or adhesive discs. Their feeding habits vary, with some species exhibiting herbivorous, carnivorous, or omnivorous diets. Reproduction occurs both sexually and asexually, depending on the species and environmental context. While their active life cycle proceeds under favorable conditions, it is their ability to enter a variety of dormant states that underpins their exceptional resilience.

One of the most extensively studied dormant states in tardigrades is cryptobiosis, a physiological condition characterized by a dramatic reduction in metabolic activity in response to adverse environmental conditions. Anhydrobiosis, a specific form of cryptobiosis induced by desiccation, involves the loss of nearly all body water and the formation of a tun, a dehydrated, barrel-shaped body form. In this state, tardigrades exhibit remarkable tolerance to desiccation, surviving prolonged periods of extreme dryness that would be lethal to most other organisms. Upon rehydration, metabolic activity resumes, and the tardigrade returns to its active state.

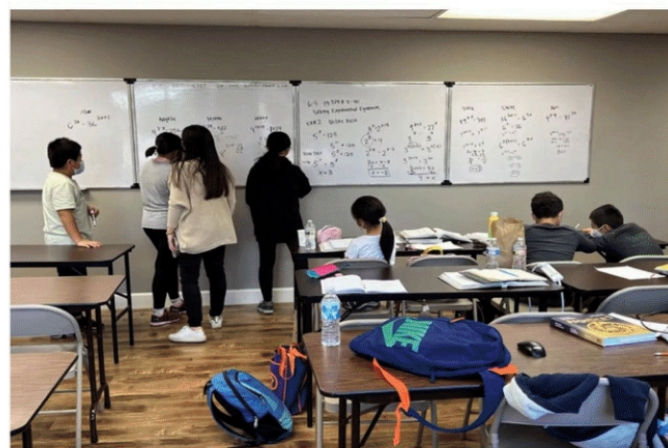
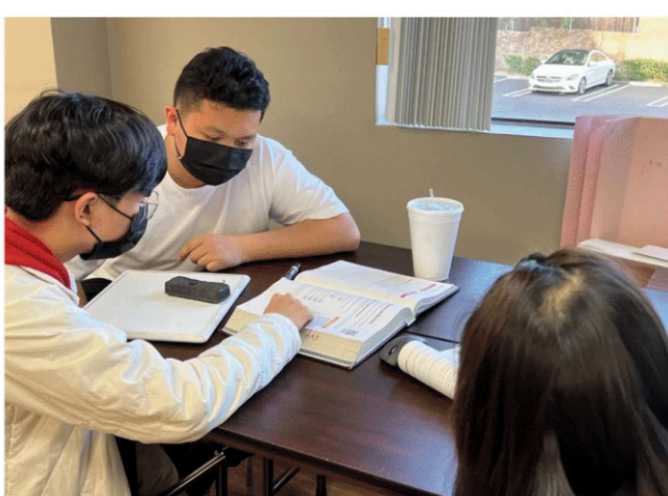
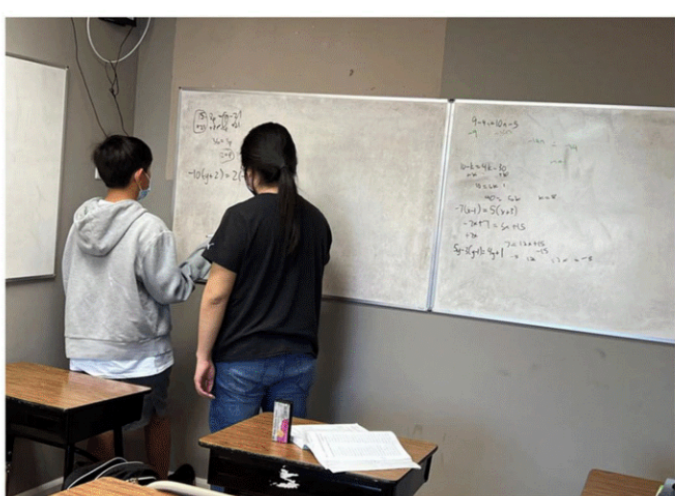
Furthermore, tardigrades have demonstrated the ability to withstand extreme temperatures, tolerating near-absolute zero temperatures and exposures to temperatures exceeding the boiling point of water for short durations. Their resistance to ionizing radiation is also notable, with some species exhibiting significantly higher tolerance levels compared to other metazoans. Additionally, tardigrades have survived exposure to the vacuum and ionizing radiation of outer space, as demonstrated by experimental studies conducted in low Earth orbit. Their capacity to endure high hydrostatic pressures, such as those found in the deep ocean, further underscores their physiological robustness.

The mechanisms underlying the exceptional stress tolerance of tardigrades are multifaceted and involve a combination of physiological and molecular adaptations. The formation of the tun during anhydrobiosis is associated with the synthesis of protectant molecules, such as trehalose, a non-reducing sugar that is thought to stabilize cellular membranes and proteins. Additionally, the production of intrinsically disordered proteins, known as Tardigrade-specific intrinsically disordered proteins (TDPs), has been implicated in providing cellular protection during desiccation and other stresses. The precise roles of these and other molecular mechanisms in mediating tardigrade resilience are areas of ongoing research.

The study of tardigrade extreme tolerance holds significant implications for various scientific disciplines. Understanding the molecular and physiological adaptations that enable these organisms to survive extreme conditions may provide insights relevant to astrobiology, informing the search for life in extraterrestrial environments characterized by harsh conditions. Furthermore, the protective mechanisms employed by tardigrades could potentially be harnessed for applications in biotechnology and biomedicine, such as the preservation of biological materials.

In conclusion, the phylum Tardigrada represents a compelling example of biological resilience. Their ability to withstand a remarkable range of environmental extremes through the utilization of various dormant states and protective mechanisms has established them as a model organism for studying the limits of life. Continued research into the unique adaptations of tardigrades promises to further our understanding of biological survival strategies and potentially yield valuable applications in diverse scientific and technological fields.





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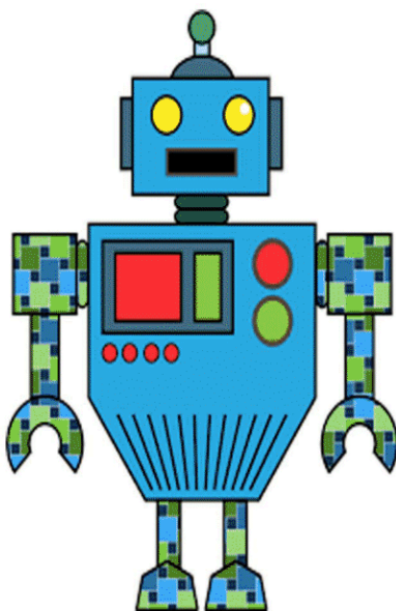
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